

Question Number	Answer	Additional Guidance	Mark
4a	A description that includes three of the following: • find a suitable concentration range for ABA (1) • find a suitable method for measuring starch digestion (1) • find a suitable {temperature/pH} (for starch digestion) (1) • find a suitable timescale (for starch digestion) (1)	Accept suitable concentrations of ABA Accept suitable method for measuring amylase production Accept species (of wheat) Accept find a suitable time for soaking endosperm in ABA (1)	3

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4b	An answer that includes eight of the following: • clear statement of the dependent variable e.g. size of clear area (1) • some description of aseptic technique e.g. rinse endosperm in {sodium hypochlorite/sterile water} (1) • soak the endosperm with ABA and place on starch agar (1) • stated time for incubation 24-72hrs (1) • method of determining clear area e.g. trace for total area or take (several) diameter measurement(s) (1) • identify one variable to be controlled and description of how this variable is controlled (1) • identify second variable to be controlled and description of how this variable is controlled (1) • repeats at each ABA concentration or repeat the whole experiment (1) • test at 5 different concentrations of ABA (1)	Accept absorbance / transmission using colourimeter Accept using disinfectant (of wheat or apparatus, but not bench) Accept place endosperm in tube containing starch solution Accept use colorimeter to measure {transmission / absorbance} (if starch in tube method used) Accept pH temperature humidity light intensity species / variety, age of seed, mass / size of seed	8

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4c	- raw data table with headings and means calculated from repeats (1) - line graph format with labelled axes (1) - use of an appropriate statistical test (1)	Accept any header with units even if incorrect Accept description of calculating mean Do not accept units in the body of the table Accept scatter graph Accept correlation test or named correlation test	3

Question Number	Answer	Additional Guidance	Mark
4d	- difficult to measure small clear areas (1) - difficult to maintain aseptic conditions (1) {growth regulators / chemicals} in seed may affect amylase production	Accept difficulty in using the colourimeter or judging an end point Accept {seeds / agar} may be contaminated with bacteria Accept genetic variation in seeds affects {amylase production / size of endosperm}	2

